

1- Give the order of the following:

a- First Ionization Energy: ${}_3\text{Li}$, ${}_7\text{N}$, ${}_9\text{F}$ and ${}_{11}\text{Na}$

F	>	N	>	Li	>	Na
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${}_3\text{Li}$: $1s^2 2s^1 \Rightarrow$ Period = 2 and Group = 1

${}_7\text{N}$: $1s^2 2s^2 2p^3 \Rightarrow$ Period = 2 and Group = 5

${}_9\text{F}$: $1s^2 2s^2 2p^5 \Rightarrow$ Period = 2 and Group = 7

${}_{11}\text{Na}$: $1s^2 2s^2 2p^6 3s^1 \Rightarrow$ Period = 3 and Group = 1

b- Electronegativity: ${}_{20}\text{Ca}$, ${}_{37}\text{Rb}$, ${}_9\text{F}$ and ${}_{15}\text{P}$

F	>	P	>	Ca	>	Rb
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${}_{20}\text{Ca}$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 \Rightarrow$ Period = 4 and Group = 2

${}_{37}\text{Rb}$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1 \Rightarrow$ Period = 5 and Group = 1

${}_9\text{F}$: $1s^2 2s^2 2p^5 \Rightarrow$ Period = 2 and Group = 7

${}_{15}\text{P}$: $1s^2 2s^2 2p^6 3s^2 3p^3 \Rightarrow$ Period = 3 and Group = 5

c- Electronic configuration: ${}_{27}\text{Co}$, ${}_{33}\text{As}$, ${}_{18}\text{Ar}$ and ${}_{14}\text{Si}$

As	>	Co	>	Ar	>	Si
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d- The entropy(S) of the gases: ${}_{54}\text{Xe}$, ${}_{18}\text{Ar}$, ${}_2\text{He}$ and ${}_{36}\text{Kr}$

Xe	>	Kr	>	Ar	>	He
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e- The frequency: 2.5×10^4 MHz, 25 GHz, 25×10^6 Hz and 250 kHz

2.5×10^4 MHz	>	25 GHz	>	25×10^6 Hz	>	250 kHz
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2.5×10^4 MHz = $2.5 \times 10^4 \times 10^6 = 25 \times 10^9$ Hz

25 GHz = 25×10^9 Hz

25×10^6 Hz

250 kHz = 250×10^3 Hz = 25×10^4 Hz

2- Give the order of the following:

a- Atomic radius: $_{11}\text{Na}$, $_{17}\text{Cl}$, $_3\text{Li}$ and $_8\text{O}$

Na	>	Li	>	Cl	>	O
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$_{11}\text{Na}$: $1s^2 2s^2 2p^6 3s^1 \Rightarrow$ Period = 3 and Group = 1

$_{17}\text{Cl}$: $1s^2 2s^2 2p^6 3s^2 3p^5 \Rightarrow$ Period = 3 and Group = 7

Cl most right element even if it is in 3th orbit below **Li** by one step but it is to the right by 6 steps

$_3\text{Li}$: $1s^2 2s^1 \Rightarrow$ Period = 2 and Group = 1

$_8\text{O}$: $1s^2 2s^2 2p^4 \Rightarrow$ Period = 2 and Group = 6

b- Electronegativity: $_{12}\text{Mg}$, $_3\text{Li}$, $_9\text{F}$ and $_8\text{O}$

F	>	O	>	Li	>	Mg
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$_{12}\text{Mg}$: $1s^2 2s^2 2p^6 3s^2 \Rightarrow$ Period = 3 and Group = 2

$_3\text{Li}$: $1s^2 2s^1 \Rightarrow$ Period = 2 and Group = 1

$_9\text{F}$: $1s^2 2s^2 2p^5 \Rightarrow$ Period = 2 and Group = 7

$_8\text{O}$: $1s^2 2s^2 2p^4 \Rightarrow$ Period = 2 and Group = 6

c- Electronic configuration: $_{25}\text{Mn}$, $_7\text{N}$, $_{10}\text{Ne}$ and $_{16}\text{S}$

Mn	>	S	>	Ne	>	N
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d- The entropy(S) of the molecules: HI, HBr, HF and HCl

HI	>	HBr	>	HCl	>	HF
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e- The volume: 20,000 μL , 2.00×10^3 mL, 200 cm^3 and 200,000 nL

2.00×10^3 mL	>	200 cm^3	>	20,000 μL	>	200,000 nL
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$$20,000 \mu\text{L} = 20000 \times 10^{-6} = 2 \times 10^{-2} \text{ L}$$

$$2.00 \times 10^3 \text{ mL} = 2 \times 10^3 \times 10^{-3} = 2 \text{ L}$$

$$200 \text{ cm}^3 = 200 \text{ mL} = 200 \times 10^{-3} = 2 \times 10^{-1} \text{ L}$$

$$200,000 \text{ nL} = 200000 \times 10^{-9} = 2 \times 10^{-4} \text{ L}$$

3- Give the order of the following:

a- Atomic radius: ${}_{7}\text{N}^{-1}$, ${}_{9}\text{F}^{-1}$, ${}_{11}\text{Na}^{+}$, and ${}_{12}\text{Mg}^{+2}$

Mg	>	${}_{7}\text{N}^{-1}$	>	${}_{9}\text{F}^{-1}$	>	${}_{11}\text{Na}^{+}$
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b- The volume: 100 mL, $1.00 \times 10^3 \text{ cm}^3$, 10,000 μL and 100,000 nL

$1.00 \times 10^3 \text{ cm}^3$	>	100 mL	>	10,000 μL	>	100,000 nL
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4- Give the order of the following:

Atomic radius: ${}_{20}\text{Ca}^{+2}$, ${}_{17}\text{Cl}^{-}$, ${}_{19}\text{K}^{+}$, and ${}_{16}\text{S}^{-2}$

S^{-2}	>	Cl^{-}	>	K^{+}	>	Ca^{+2}
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b- The mass: 70,00 μg , $7.00 \times 10^3 \text{ mg}$, 70,000 nL and 7 kg

7 kg	>	$7.00 \times 10^3 \text{ mg}$	>	70,00 μg	>	70,000 nL
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Exercise 1: Give the order of the following:

a- Atomic radius: ${}_{13}\text{Al}$, ${}_{8}\text{O}^{-2}$, ${}_{19}\text{K}^{+}$, and ${}_{17}\text{Cl}$

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b- Atomic radius: ${}_{13}\text{Al}^{+3}$, ${}_{8}\text{O}^{-2}$, ${}_{19}\text{K}^{+}$, and ${}_{17}\text{Cl}^{-1}$

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